

Entity Resolution: published research vs. the upstream pipeline

Gap analysis, prioritised findings and concrete fixes for **Ruthwik000/amazonsummerML** as adapted in `colab_upstream_cell.txt`

Field	Detail
Scope	Methodology comparison only. No challenge dataset, ground truth, leaderboard or third-party solution to this challenge was searched for or used.
Under review	Ruthwik000/amazonsummerML @ 8445b7f (read-only clone in <code>_upstream/</code>), 22 source files, <code>run_all.sh</code> drives 12 steps.
Research corpus	SIGMOD Programming Contests 2021/2022, ESWC 2024, NAACL 2024, VLDB 2021, plus record-linkage literature. Full survey: <code>RESEARCH_entity_resolution_methodology.md</code> .
Verdict	Upstream is already competition-grade. The highest-return work is not a better model - it is France coverage, validation honesty, and packaging.

REVISED EDITION - every finding in this report has now been re-tested

The first edition shipped with **untested** findings. Given that its two top-ranked findings both turned out to be false on measurement, the remaining six were re-checked against the code and data rather than republished as-is. **Four of eight findings are now withdrawn or partly withdrawn: F1** (France postal codes) and **F2** (FR_REGIONS) are false; **F6** is half wrong - mutual exclusion *does* exist at `crossfit.py:37-44`, only the missing per-entity cap stands; **F8** Gaps 3 and 4 are both resolved, Gap 3 never was a bug. **F3, F4, F5 and F7** are confirmed and unchanged. The full measured record, including three further France hypotheses that were also ruled out, is in `FRANCE_FINDINGS.md`. **Standing lesson: a claim ships with the script that produced it, or it does not ship.**

Severity has been re-ranked, and the ranking changed

In the first edition F1 and F2 were rated CRITICAL on the reasoning that they affected France, the only scored country, at a cost of a line each. Both are false, so that entire justification collapses. **The highest-confidence remaining work is F8 Gap 1** - packaging and the methodology document - which is a reporting requirement, not a modelling one. F6's per-entity cap is the only live modelling change with a clear mechanism behind it. Read the priority table in section 6 as superseding the original ordering.

Correction to a working assumption carried in earlier notes

Earlier notes stated that "a smaller candidate set ranks higher, beyond the model score." **The brief does not say that.** It says `candidate_pairs.tsv` "*is not scored on the leaderboard*" and is used "to analyse blocking quality (recall ceiling, reduction ratio) and to verify your pipeline." The only leaderboard-scored artefact is `matching_results.tsv`.

Consequence: shrinking the candidate set buys no leaderboard points; it only moves the recall ceiling you are reviewed on. A research sub-agent was briefed with this wrong premise and repeated it - that claim is retracted here and re-derived from the primary sources below.

1. What the upstream pipeline actually does

The design under review is stated precisely first, with file and line references. This matters because most findings below are about *which population a number applies to*, not about gross architectural errors.

- **Blocking (retrieval).** Composite keys over 5 kinds - name x address, name-pair, address-number x address-word, full sorted name, single name token - hashed to u64 and scoped by country. Scored by summed IDF of shared keys, top 50 kept, plus a relative floor (rel=0.2). *src/keys.py, src/blocking.py:20-39, src/run_blocking.py:8.*
- **Stage 1 - learned candidate ranker.** Two LightGBM models, each trained on one half of the queries and scoring the other half, giving every training pair an out-of-fold score. *src/crossfit.py:60-100.*
- **Stage 2 - acceptance model.** LightGBM over each query's best candidate with cluster context and query-to-query similarity; threshold tuned for macro F0.5 on V. *src/crossfit.py:110-161.*
- **Assembly.** *matching_results.tsv* = accepted pairs; *candidate_pairs.tsv* = exactly the set stage 2 scores, filtered at $P1_MIN = 0.02$. *src/make_submission.py:68-77.*

Reported funnel: **30.5** candidates per Source-1 entity after retrieval, **3.97** after the learned ranker (6.9M pairs). A full run is quoted at 3-4 hours on 2 cores with peak under 6 GB, because every step is chunked per country or per query range. This is well engineered, and the two-stage retrieve-then-rank shape is the same one the literature converges on.

Stage	Cand. / S1	Role	Reference
Raw key retrieval	30.5	Recall ceiling	<i>run_blocking.py:27</i>
Learned ranker (stage 1)	3.97	What stage 2 scores = <i>candidate_pairs.tsv</i>	<i>make_submission.py:74</i>
Stage 2 + threshold THR	accepted	<i>matching_results.tsv</i>	<i>make_submission.py:68</i>

Table 1 - The candidate funnel as instrumented in the upstream code.

2. The research corpus, and what each source actually claims

The SIGMOD Programming Contests are the two most valuable sources, because one of them scores blocking almost exactly the way this challenge reviews it. Confidence tags are carried through from the underlying survey, which marks each claim [VERIFIED] / [INFERENCE] / [THIN] / [GAP].

Source	What it contributes	Bearing on this task
SIGMOD 2022 - Blocking System for ER (Chu Data Lab / DBGroup UNIMORE); winner Brinkmann & Peeters, U. Mannheim	Task was <i>only</i> blocking. Ranked on average recall at a hard candidate-set cap , ties broken by runtime. 1M rows/dataset, 1M + 2M pair budgets, 35 min CPU-only.	Closest structural analogue in public competition. Confirms candidate size is a genuine engineering constraint, but not a leaderboard score.
SC-Block (ESWC 2024), Brinkmann, Shraga, Bizer - arXiv:2303.03132	Supervised contrastive learning places records in an embedding space, then ANN retrieval. Smaller candidate sets than 8 baselines; pipelines 1.5-2x faster, 8x on a large benchmark (2.5h -> 18min against 5 min training). Benchmarks at 99.5% pair completeness .	The learned-retriever upgrade path. Its operating point is far more generous than this task needs.
PLMs for Entity Blocking (NAACL 2024), Wang & Zhang - 2024.naacl-long.483	Evaluates SOTA blockers <i>and</i> neural-IR models, studying in-distribution vs out-of-distribution generalisation - explicitly called out as under-studied yet important in real applications.	Directly on point for the unseen-third-country risk. Reads as a caution, not a recipe.
Ditto (VLDB 2021), Li, Li, Suhara, Doan, Tan - arXiv:2004.00584	PLM fine-tuned as sequence-pair classification. Up to +29% F1 over prior SOTA, +9.8% more from input highlighting, long-value summarisation and augmentation. Reaches prior SOTA with at most half the labelled data. 789K x 412K company records, F1 96.5%.	Precedent for a company-record matcher, and for label efficiency mattering. Needs an MIT/Apache model under 8B - permitted, but costly.

Source	What it contributes	Bearing on this task
Bias in error estimation (BMC Bioinformatics 2006), Varma & Simon - doi:10.1186/1471-2105-7-91	CV error for the optimally-tuned classifier was substantially biased; on null datasets the tuned SVM estimate was <30% true error 38% of the time , with test performance no better than chance. Nested CV "gives an estimate very close to that obtained on the independent testing set."	The direct justification for finding F3 . Threshold choice is a tuning step by any reasonable definition.
Self-map audit of this repo - audit_normalize.py	Counts parsed from the real normaliser, not recalled. FR_REGIONS: 14 entries, 4 distinct values . US_STATES: 52/52. IN_STATES: 49/37.	Quantifies F1 and F2 .

Table 2 - Primary sources and what each actually claims.

3. Gap matrix: upstream vs. the literature

Technique from the literature	Upstream status	Verdict
Two-stage retrieve-then-rank over all attributes	Present and well built (30.5 -> 3.97)	Already done. No action.
Learned / embedding retriever (SC-Block class)	Absent - retrieval is IDF over hashed keys	Real gap, but see F7 - not the first move .
Country-agnostic normalisation with fallbacks for an unseen country	Structurally safe (country-scoped keys, .get fallback); France tables verified sufficient on this dataset	No gap. F1/F2 were refuted on measurement; six further audits agree. Do not extend the tables.
Abstention-aware thresholding for singletons	Single global THR tuned on V	Partially present; see F5 and F6.
Nested / cross-fitted threshold selection	V is used for early stopping, thresholding <i>and</i> reporting	Gap - F3. Affects reported honesty, not the submission.
Set-level constraints: cluster-size cap, mutual exclusion	Mutual exclusion present (crossfit.py:37-44 anti-join); per-entity list uncapped	Partial gap - F6. Only the cap is missing. The exclusion half was a misreading.
Metric-exact local scorer including singleton semantics	Present; verified against sklearn fbeta_score over 400 randomised cases	Already done. Keep it.
Fellegi-Sunter / m-u match weights as calibrated features	Not present (raw similarity features instead)	Optional; the learned ranker largely subsumes it. Low priority.
Candidate recall@budget measured explicitly	Funnel sizes reported, but no recall-ceiling measurement	Gap - F5. Cheap to add on V.

Table 3 - Most high-value literature techniques are already in place, or are gaps with unclear payoff. The two exceptions are France coverage and validation honesty.

4. Findings and fixes, in priority order

Severity reflects effect on the private-leaderboard score for the *French* test set, divided by cost. **This ordering is the revised one.** In the first edition F1 and F2 dominated the list on the reasoning that they were cheap and France-specific; both were then measured and found false, so the top of the ranking has changed. Items marked RESOLVED or REFUTED are retained for auditability, not as work items.

F1

[REFUTED - DO NOT ACT ON THIS] France postal codes are never captured

Status	This finding is FALSE and has been withdrawn. It was produced by reading <code>normalize.py</code> without measuring the data. The original text is retained below only so the error is auditable.
Where	<code>src/normalize.py:336-338</code>
Evidence	<code>if len(n) == 6 and country == "India": pin = n</code> A parse of the normaliser confirms <code>pin</code> is assigned in exactly one place, under an India-only guard. Verified by <code>grep</code> across <code>prep.py</code> and <code>normalize.py</code> .
Consequence	US ZIP codes (5 digits) and French codes postaux (5 digits) are never stored in <code>pin</code> . That field feeds a blocking key (<code>keys.py:44</code>) and a ranker feature. Both are therefore constant/dead for France - and France is the entire scored test set. The strongest cheap locality signal in a French address is simply not being used.
Fix	Widen the guard. Capture 5-digit codes for US and France in addition to the existing 6-digit Indian PIN. Optionally also capture the 2-character French departement forms (2A, 2B) which do not parse as integers.
Effort	~1 line plus regression tests.
Risk	Low, but re-run blocking: adding a new key kind changes the candidate set and invalidates cached stage-1 outputs. Do it before any long run, not after.

F2

[REFUTED - DO NOT ACT ON THIS] FR_REGIONS covers 4 of 18 regions and is excluded from the abbreviation self-map

Status	This finding is FALSE as a data-quality claim and has been withdrawn. Replicating <code>normalize_address</code> 's exact component matching, state resolves for 100.00% of France rows (259,452/259,452, zero unmatched). The table is sufficient because this dataset uses only three modern regions. The self-map defect at line 252 is real code but has no measurable effect here. The original text is retained below only for audit.
Where	<code>src/normalize.py:243-254</code>
Evidence	<code>FR_REGIONS</code> has 14 entries mapping to only 4 distinct values (<code>hdf</code> , <code>naq</code> , <code>pdl</code> , <code>idf</code>) - versus <code>US_STATES</code> 52/52 and <code>IN_STATES</code> 49/37. It omits Bretagne, Normandie, Grand Est, Occitanie, AuRA, PACA, Centre-Val de Loire, Bourgogne-Franche-Comte and Corse, plus 90 of 101 departements.
Second defect	The self-map loop reads for <code>_m in (US_STATES, IN_STATES):</code> at line 252 - FR_REGIONS is not in the tuple . An already-abbreviated value such as <code>idf</code> is therefore not recognised as a state and falls through into the address token list, adding noise.
Fix	Populate all 18 regions and the 101 departement codes; add <code>FR_REGIONS</code> to the self-map loop. This is a pure data-table edit with no structural change.
Why it matters	The state feature and the state match in stage 2 are close to useless for France as written, so the model falls back on weaker string similarity for a third of the scored rows.

Effort	Data entry, ~1 hour. Highest certainty-to-effort ratio in this report.
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F3

V is used for early stopping, threshold selection AND reporting - val_f05 is optimistic

Where	src/crossfit.py:84 (early stop on V), 153-156 (threshold sweep on V), 161 (reports best_thr[0])
Evidence	The same V split (3% of Source-1 entities) serves three roles at once. A controlled simulation in this workspace put single-step optimism at roughly +2.4% .
Literature	Varma & Simon: CV error for the optimally-tuned classifier is substantially biased, and on null data the tuned estimate was <30% true error 38% of the time. Their prescription is that <i>all</i> steps including parameter tuning be repeated inside each CV loop.
Important nuance	Tuning the shipped threshold on V is legitimate - that is what a validation set is for. Only the <i>reported</i> number is compromised. This is a reporting-integrity fix, not a submission change.
Fix	Split V in two by hash: V-tune for early stopping and threshold selection, V-report untouched for the headline figure, and report both. Then add a second independent holdout - the current design has none.
Effect on score	None on the submission. Large effect on how credible the methodology document is in review - and the package <i>is</i> reviewed before final rankings are confirmed.

F4

The Vkeys evaluation population is narrower than the test-time population

Where	src/crossfit.py:124
Evidence	Vkeys = best.join(qV, on="rid").join(s1V, on="s1") restricts both the query and the Source-1 side to V.
Consequence	The macro F0.5 estimate is computed over a sub-population in which cross-links between V and non-V entities cannot appear. At test time any query may match any Source-1 entity, so the real macro average is taken over a broader - and on these grounds slightly different - population.
Fix	Evaluate V queries against <i>all</i> their candidate Source-1 entities, not only those whose s1 is in V. The gold-truth join already uses the full truth, so this is a change to the prediction filter, not to the scorer.
Effort	Small, but it requires care to keep the scorer macro semantics intact.

F5

Candidate set is top-1 per query with a very loose floor - measure the recall ceiling before touching it

Where	src/make_submission.py:74
Evidence	cand = best.filter(pl.col("p1") >= P1_MIN) - best is stage 2's input, one candidate per query, so the candidate set is exactly {best candidate : p1 >= 0.02}.
Contract check	This is correct against the brief : candidates must be the last stage's input, and every ID in matching_results.tsv must appear in it. Since acc filters a subset of cand on the same P1_MIN (line 68), the subset invariant holds. No bug here.

The real issue	Recall ceiling is capped at top-1 by construction. The SIGMOD 2022 winners instead kept a <i>ranked</i> list and filled to an explicit pair budget, which decouples 'what the model scores' from 'what the reviewer audits'.
Fix	Measure the recall ceiling on V as a function of k and of the p1 floor, then emit a ranked top-k (k = 3-5) above a higher floor. Because candidate_pairs.tsv is not leaderboard-scored , this costs no score and raises the recall ceiling visible in review.
Local fallback evidence	The measured cap/recall tradeoff on the local pipeline was 4.89 cand/S1 at 0.4108 gold recall (cap 5) rising to 9.63 at 0.4652 (cap 10). Upstream's learned ranker reaches 3.97, a materially better point on the same curve.

F6

[PARTLY REFUTED] Per-entity cap is a real gap; the mutual-exclusion half is an open question

Status	Half of this finding is wrong. The original text asserted there is <i>no mutual exclusion</i> . There is: <code>home_table()</code> at <code>crossfit.py:37-44</code> anti-joins each priority tier against the tiers already assigned (<code>r.join(seen, on="rid", how="anti")</code>), so a query is assigned to exactly one home partition and cannot be scored twice. The per-entity cap half stands and is worth acting on.
Where	<code>src/make_submission.py:69-70</code> (cap absent) and <code>src/crossfit.py:37-44</code> (exclusion present)
Still valid - no cap	<code>s1.join(acc.group_by("s1").agg(pl.col("entity_id").sort().str.join(",").alias("matched_entity_ids")))</code> - every accepted ID is concatenated with no cap.
Why it costs precision	With F0.5, a false merge costs roughly 4x a missed link, and singletons score 0.0 for <i>any</i> prediction. A chaining error that absorbs 10 wrong records into one entity costs far more than one that absorbs 2. A per-entity cap of the k best by p2 bounds that damage directly.
Still open	Whether one S2/S3 record may be claimed by two different S1 entities. The brief rejects duplicates <i>within</i> a list but is less explicit about cross-entity claims. Verify against <code>utils/validate_submission.py</code> before assuming it is either allowed or forbidden.
Fix	Cap <code>matched_entity_ids</code> per S1 at the top k by p2. Treat cross-entity collision resolution as a separate, separate-lookup question - do not bundle it with the cap.
Effort	~5 lines, plus a decision on the collision question.

F7

A learned / embedding retriever is a real gap but explicitly NOT the next move

The technique	SC-Block and the SIGMOD 2022 winner: fine-tune a small distilled Transformer with a supervised contrastive loss, embed records, retrieve by approximate nearest neighbour (FAISS), rerank with a symbol-level similarity such as Jaccard, and fill to a pair budget.
Reported	Smaller candidate sets than eight baselines; pipelines 1.5-2x faster, 8x on a large benchmark (2.5h -> 18 min) against 5 min of training.
Why not now	Three reasons. (1) Out-of-distribution risk : the NAACL 2024 study shows SOTA blockers' generalisation to unseen data is under-studied and a real concern - a contrastive retriever trained on US+India faces precisely that, on the only country that scores. (2) Wrong operating point : SC-Block benchmarks at 99.5% pair completeness, far more generous than a precision-weighted metric needs. (3) Prohibited ingredient : the 2022 winner's pretraining set came from Common Crawl via schema.org - disqualifying here.
Revisit only if	F1-F6 are done and a second-country holdout exists to measure the OOD drop honestly. Without that holdout you cannot tell an improvement from a regression.

Verdict	High cost, uncertain payoff, and it would consume the effort that F1 and F2 - a few lines each - return with high confidence.
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F8

Packaging and reproducibility gaps that can cost the submission outright

Brief requirement	The final package is <team>_submission.zip containing output/ (both TSVs), code/business_entity_resolution/ with src/, README.md and pinned requirements.txt, plus a filled Documentation_template.md.
Gap 1	The upstream repo has src/ at top level and no Documentation_template.md. It needs restructuring to the required path, and the methodology document has to be written - the brief has no page limit and explicitly prioritises depth.
Gap 2	During the challenge the portal takes matching_results.tsv only ; the zip is the end-of-challenge package. Confirm which the portal expects per upload rather than assuming.
Gap 3 - RESOLVED	<i>Originally reported here:</i> the cell ran <code>pip install -r requirements.txt</code> before cloning, so a clean run would not find the file. Re-checked, and this is not a bug. The clone is unconditional, in its own step (<code>colab_upstream_cell.txt:50-53</code> , <code>subprocess.run(["git", "clone", ...], check=True)</code>), and the install line falls back to explicit package names when the requirements file is absent. Withdrawn.
Gap 4 - RESOLVED	<code>check_cell.py</code> originally compared only script names, step order and GT placement. It now performs a full argument-level diff against <code>run_all.sh</code> (12/12 exact match), so a silent flag drift cannot ship.
Still open	Gaps 1 and 2 only. Gap 1 is real packaging work and the largest remaining item; Gap 2 is a one-question check against the upload portal.

5. What not to do

Tempting move	Why it is wrong here
Port the SIGMOD 2022 winner end to end, including its Common Crawl pretraining	External data augmentation is explicitly prohibited ; the penalty is immediate disqualification. Transfer the idea, not the corpus.
Shrink candidate_pairs.tsv to chase a smaller number	It is not leaderboard-scored. You would trade away the recall ceiling for a metric that does not exist, and the ceiling is what gets reviewed.
Hard-code {US, India} or filter out unknown countries	The brief calls country an open set and warns against exactly this. Upstream is already correct (country-scoped keys, STATE_MAPS.get); fix the <i>data</i> in F2, not the code path.
Quote the 0.98612 val_f05 as a headline in the methodology document	It is the tuned-on-V figure (F3). Quoting it invites exactly the audit it will not survive, in a package reviewed before final rankings are confirmed.
Use a geocoding or business-registry API to enrich addresses	Prohibited. The absence of a general-purpose geocoder from upstream is correct and must be preserved.
Keep designing against the 'smaller candidate set ranks higher' premise	Retracted - a misreading of the brief (cover callout). It has already propagated into one research sub-agent's output; do not rely on that output without re-checking it.

Table 5 - Traps, each traced to a specific source line or brief clause.

6. Recommended order of work

#	Action	Findings	Cost	Affects score?
1	Restructure the repo to the required package layout; write the methodology doc	F8	Hours	Submission validity
2	Cap matched_entity_ids per S1 at the top k by p2 (mutual exclusion already exists)	F6	~5 lines	Yes - precision
3	Split V into tune/report halves; add a second untouched holdout	F3, F4	Small	No - reporting honesty
4	Measure candidate recall ceiling on V for k = 1..5; emit a ranked top-k	F5	Small	No (review only)
5	Confirm with the portal whether the zip or matching_results.tsv is uploaded per attempt	F8 G2	Minutes	Submission validity
6	Only then consider a learned retriever, and only with a second-country holdout	F7	Days	Unknown - measure first
-	[REFUTED] Widen the postal-code guard	F1	-	No - do not do
-	[REFUTED] Complete FR_REGIONS / add to self-map loop	F2	-	No - do not do
-	[RESOLVED] Reorder clone before pip install	F8 G3	-	Done - was never a bug

Table 6 - Sequencing, revised after re-measurement. The first edition opened with two one-line France fixes; both are now known to be no-ops, so **the highest-confidence remaining work is packaging and the methodology document**, which the brief requires and which is reviewed before final rankings are confirmed. The only live modelling change with a clear mechanism behind it is the F6 per-entity cap. Items 3-4 protect the submission's credibility; item 6 stays last until there is a second-country holdout to measure an OOD drop honestly.

Bottom line

The upstream pipeline is genuinely good engineering, and most of what the literature recommends is already in it. **The revised conclusion is the opposite of the first edition's on France.** That edition argued the gap was concentrated in France - specifically its postal code and region table - and rated both CRITICAL. Both were then measured and found to be no-ops: France carries house numbers rather than postcodes, and its three regions in use are all already mapped. Six further France-specific audits returned the same verdict, including an independent convergence showing that completing FR_REGIONS would actively *delete* live IDF feature mass. **What actually remains is smaller and less glamorous:** cap the per-entity output list, stop grading the validation set on itself, and produce the packaging and methodology the brief requires. That is a better use of the remaining time than any model change - and it is what the evidence supports, which is not the same as what the first edition claimed.

7. Verification commands used in this review

```
# The scripts that REFUTE F1 and F2 (these are the ones that matter now)
.venv-pdf\Scripts\python.exe check_fr_postcode.py      # F1: France dig5/row = 0.004
.venv-pdf\Scripts\python.exe check_fr_exact.py         # F2: 259452/259452 resolve = 100%
.venv-pdf\Scripts\python.exe check_pin_impact.py       # F1: India dig6/row ~ 0.000
.venv-pdf\Scripts\python.exe check_fr_regions.py      # F2: which regions actually occur

# F6: confirm mutual exclusion exists (it does - crossfit.py:37-44)
Select-String -Path _upstream\src\crossfit.py -Pattern 'anti'

# F8 Gap 3: confirm the clone is unconditional, so pip ordering is a non-issue
Select-String -Path colab_upstream_cell.txt -Pattern 'clone'

# Metric correctness, already verified in the first edition
.venv-pdf\Scripts\python.exe verify_upstream_metric.py # 400 cases vs sklearn
Full methodology survey with per-claim confidence tags: RESEARCH_entity_resolution_methodology.md. The upstream tree was
left read-only throughout; git status clean at 8445b7f.
```